

**Problem 1 (The Rayleigh Radar, 10 Points):** Given a uniform random variable  $U \sim \mathcal{U}([0, 1])$ , derive the inverse transform sampling function required to generate samples from a Rayleigh distribution with parameter  $\sigma$ :

$$f(x) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right), \quad x \geq 0.$$

**Problem 2 (Bounding the Half-Normal, 20 Points):** Suppose we want to sample from a standard Normal distribution, but only the positive half (a truncated normal). The target density

$$f(x) \propto \exp(-x^2/2), \quad x \geq 0.$$

We will use an exponential distribution

$$g(x) = \lambda \exp(-\lambda x), \quad x \geq 0$$

as our proposal distribution.

- i) (10 Points) Find the optimal scaling constant  $M$  as a function of  $\lambda$  such that  $f(x) \leq M g(x)$  for all  $x \geq 0$ .
- ii) (10 Points) Analytically determine the optimal rate parameter  $\lambda$  that minimizes the rejection rate.

**Problem 3 (The Rare Event Crisis, 15 Points):** We want to estimate the probability of a standard normal random variable exceeding a threshold of 4:

$$\mathbb{P}(X > 4) = \int_4^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx.$$

- i) (5 Points) If we use standard Monte Carlo by drawing  $N = 10^5$  samples from  $X \sim \mathcal{N}(0, 1)$ , what is the expected number of samples that will fall in the region  $X > 4$ ?
- ii) (10 Points) Propose a shifted Gaussian importance sampling distribution

$$q(x) = \mathcal{N}(\mu, \sigma^2)$$

that would significantly reduce the variance of this estimate. Justify your choice of  $\mu$  and write out the final Importance Sampling estimator equation.

**Problem 4 (The Bimodal Trap, 15 Points):** Let the target distribution be an unnormalized symmetric bimodal distribution:

$$\pi(x) \propto \exp\left(-\frac{(x-3)^2}{2}\right) + \exp\left(-\frac{(x+3)^2}{2}\right).$$

We will use a Metropolis-Hastings algorithm with a Gaussian random walk proposal:

$$p(x, x') = \mathcal{N}(x'; x, \sigma_p^2).$$

Assume the current state of the chain is  $x_t = 3.0$  (sitting precisely on the right peak).

- i) (5 Points) Calculate the exact acceptance ratio  $\alpha(x_t, x')$  if the proposal draws a new state  $x' = 0.0$  (the deep valley between the peaks).
- ii) (10 Points) Conceptually, what happens to the Markov Chain if the proposal variance  $\sigma_p^2$  is set extremely small (e.g., 0.01)? What happens if it is set extremely large (e.g., 100)?